

Analysis First-Year Exam, August 2024, Part 1

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let $f : [0, 1] \rightarrow [0, 1]$ be (weakly) increasing. Explain why the set

$$A = \{t \in [0, 1] : t \leq f(t)\}$$

has a supremum. Prove that the function f has a fixed point.

2. Let A be a dense subset of the metric space M . Prove that M is complete if every Cauchy sequence in A converges in M .
3. Let $f : X \rightarrow Y$ be an isometry (a distance-preserving map) between metric spaces. In each of the following cases, decide whether the image $f(X)$ must be closed in Y :
- (i) X is compact;
 - (ii) X is complete.
4. Does there exist an injective continuous real-valued function on the open unit square $U = (0, 1) \times (0, 1)$? If so, give an example; if no, prove that none exists.
5. The function $f : (0, 1) \rightarrow \mathbb{R}$ is differentiable, with bounded derivative. Must $\lim_{t \rightarrow 0+} f(t)$ exist in $\mathbb{R}$? Proof or counterexample.